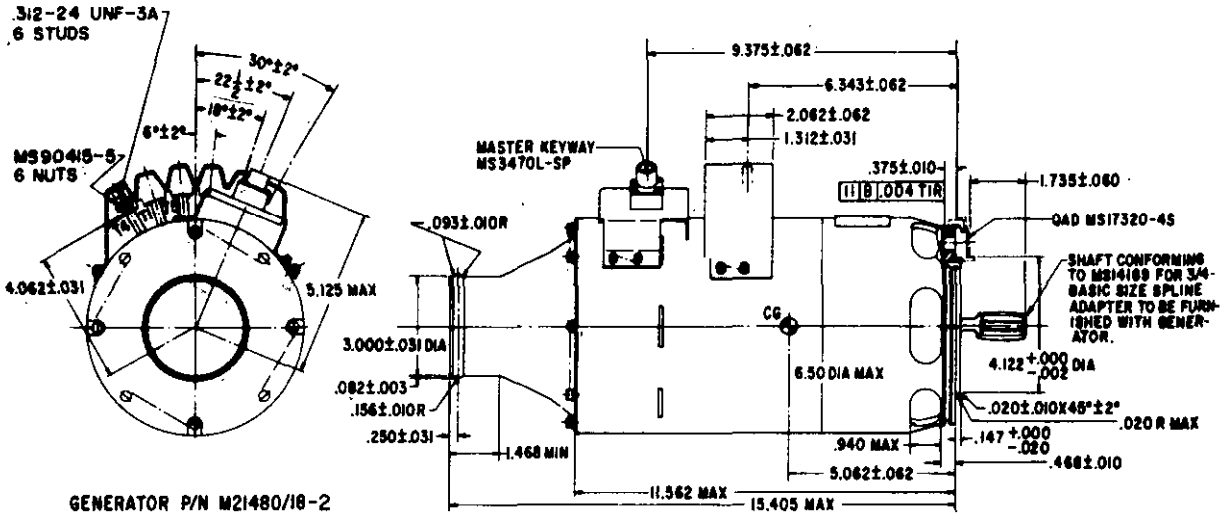
MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 30 KVA, 400 HERTZ
ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of MIL-G-21480.



WEIGHTS:
GENERATOR 53 LBS MAX

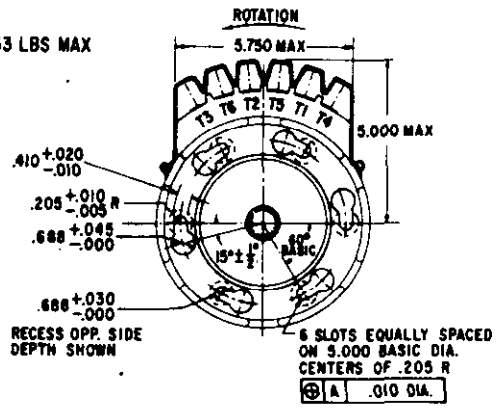
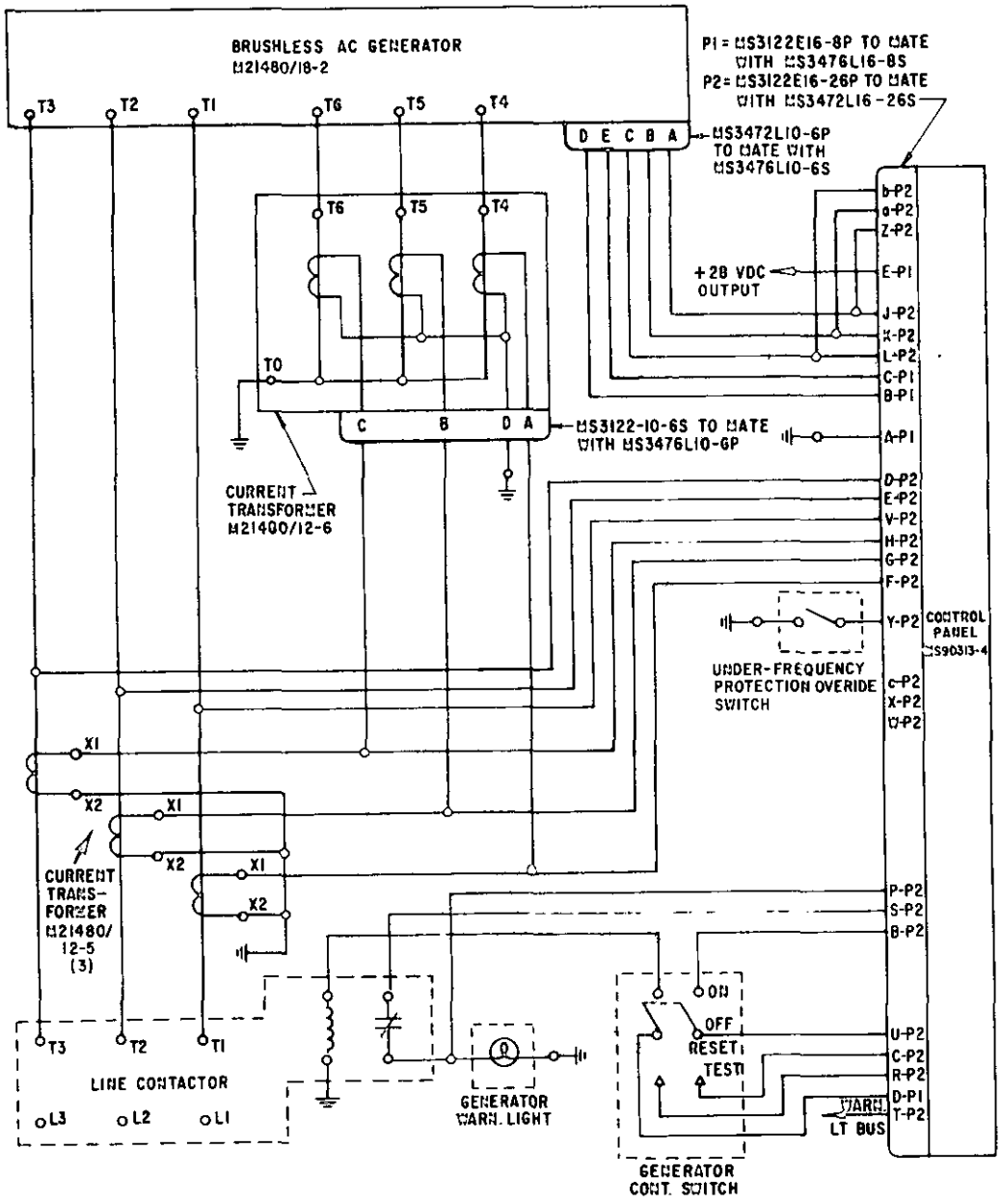


FIGURE 1

FSC 6115

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WIRING SCHEMATIC

FIGURE 2

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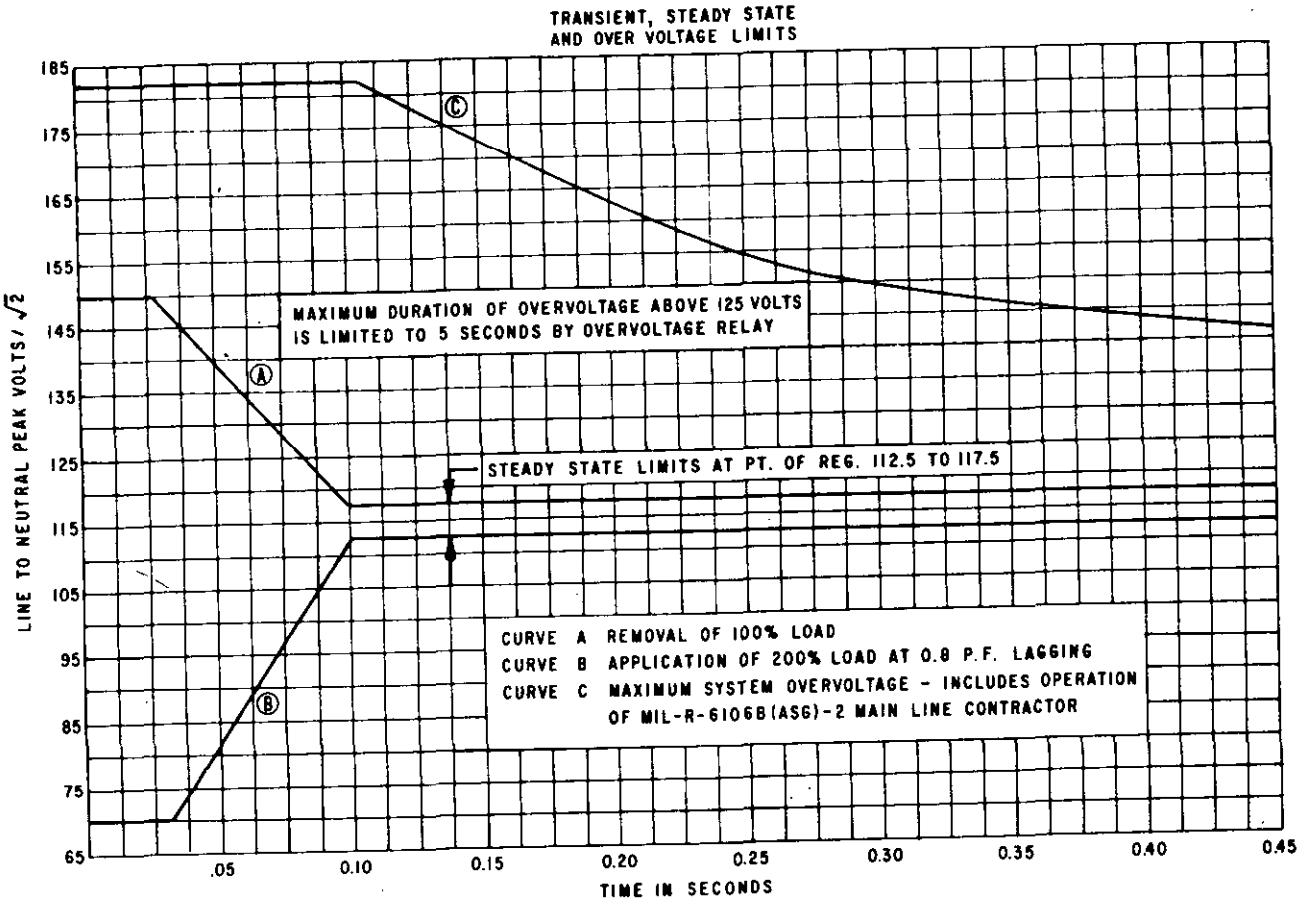


FIGURE 3

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REQUIREMENTS:

1. All paragraphs of MIL-G-21480 will apply except for the following deletions, additions or substitutions:

DELETE: Paragraphs 3.4.10.5, 3.4.10.6, 3.4.13.1, 3.4.15, 3.4.16 through 3.4.16.5, 3.5.7.5, 4.5.32, and 4.5.33 through 4.5.33.2

3.4.8 SYSTEM CAPACITY (RATING) - Delete and substitute: The rating of the system shall be as shown on Table I of the specification sheets. The nominal system rating shall be the same as the generator rating.

3.4.8.2 OVERLOAD CAPACITY - Change 5 minutes to 2 minutes.

3.4.9.3 ELECTRIC CONNECTIONS - Add: The system shall be supplied with terminals and connectors as shown in Figure 1 of the specification sheet.

3.4.9.6.1.1 ALTITUDE RATING CHART - Delete and substitute: The nominal system ratings hall be 30 KVA and values of generator maxiumum continuous load from sea level to 25,000 feet shall be as shown on Table II of the specification sheet.

3.4.10 CONTROL COMPONENT DESIGN - Delete first sentence and substitute: Control components shall be as called out on the specification sheet. The generator system interconnection wiring schematic is shown in Figure 2 of the specification sheet.

3.4.11 COLOR - Add: The color of the current transformer is optional.

3.4.13 IDENTIFICATION OF PRODUCT - Delete "Assemblies and Parts" from first sentence.

3.5.1 VOLTAGE LIMITS - Delete: "Figure 3" and substitute: Figure 3 of the specification sheet.

3.5.5 INTEGRAL CONTROL POWER - Delete and substitute: The integral control power shall be capable of supplying a continuous output of 3 amperes at 28 volts dc (nominal) to the line contactors. This power shall have the following characteristics: Minimum Volts DC; 20, Maximum Volts DC 30, Ripple; 5% maximum, $\left(\frac{\text{RMS Value of Ripple Voltage}}{\text{Average Value of D.C. Voltage}}\right)$, at rated current output, 3 amps Minimum Continuous Capacity.

3.5.7.2 OVERVOLTAGE - Delete and substitute: Overvoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and de-energize the generator before the voltage at the point of regulation exceeds the limits defined by the maximum overvoltage curve of Figure 3 of the specification sheet. Manual reset of this protective element shall be possible by putting the generator switch in the "OFF RESET" position (then to "ON") if the output voltage of the system is less than that defined by the maximum overvoltage curve of Figure 3 of the specification sheet. There shall be no automatic reset of this protective element.

3.5.7.3 UNDERVOLTAGE - Delete and substitute: Undervoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and de-energize the generator within a period of from 5 to 6.5 seconds after any phase voltage (line-to neutral) falls below 90 volts. Manual reset of this protective element shall be possible by putting the generator switch in the "OFF RESET" position (then to "ON") if the voltage on each phase exceeds 105 volts. Reset shall be automatic when the system is shut down. After either automatic or manual reset, the undervoltage protection element shall provide for the system to be automatically reconnected to the load bus within two seconds after the lowest phase voltage exceeds 105 volts.

3.5.7.4 UNDERFREQUENCY - Delete and substitute: Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus within a period of 2 to 5 seconds after the frequency falls below 370 Hz. Manual reset shall be possible by putting the generator switch in the "OFF RESET" position if the output

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frequency of the system exceeds 378 ± 3 Hz. The spread between pick-up and dropout shall not be less than 4 Hz. After either automatic or manual reset the underfrequency protection shall provide for the system to be automatically reconnected to the bus within 2 seconds after the frequency exceeds 378 ± 3 Hz. Closing the underfrequency protection override switch shall disable the underfrequency protection device.

3.5.7.6 FEEDER FAULT - Delete and substitute: The system shall provide for the generator to be de-energized and disconnected from the load bus when the fault current on any phase exceeds 18 ± 3 amperes.

3.5.8 ELECTROMAGNETIC INTERFERENCE - Delete and substitute: The conducted radio interference measurements shall be made per MIL-I-6181. The conducted radio interference produced by the system at rated current and voltage and measured between 2 output terminals or any output terminal and ground shall not exceed 5,000 microvolts over the range of 0.15 to 20 megahertz. Radiated radio interference measurements and limits shall be as specified in MIL-I-6181.

4.5.12 ELECTOMAGNETIC INTERFERENCE - Delete and substitute: The measurements and limits of paragraph 3.5.8 as amended by the specification sheet shall apply.

4.5.23.1.2 CONTROL COMPONENTS - Delete requirements above 500 Hz.

2. The generator is to meet the requirements of this specification sheet when connected as shown in Figure 2 to the following components:

   a. The control panel shall meet the requirements of MS90313, PN MS90313-4.
   b. The generator end current transformer shall meet the requirements of M21480/12-6.
   c. The load end current transformers shall meet the requirements of M21480/12-5.

3. Self Cooling required. (No external blast air provided.)

4. The system minimum time between maintenance actions shall be not less than 1200 hours. This figure shall apply to failures which will cause system operation to deviate from performance specification.

5. Spline design shall be in accordance with MS14169 3/4 inch.

6. Attaching device shall be in accordance with MS17320-4S.

NOTES:

1. Dimensions in inches. Tolerances: ±.03 inch unless otherwise specified.

2. All outline dimensions are limiting dimensions only.

3. Generator control switch, line contactor, and warning light shown for interconnection information only.

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TABLE I

GENERATOR DATA	
Rated Voltage	120/206V
Rated Output	30 KVA (Nominal)
Phase	3
Frequency	380-420 Hz
Speed Range	7600-8400 RPM
Max. Speed for Regulation	10,000 RPM
Overspeed	11,000 RPM
Rated Power Factor	0.75 Lag to 1.0
Efficiency Min. at Rated Load	87%
Max. Weight	53 Lbs.
Overhung Moment-Max.	280 In. - Lb.
Class	C
Shear	1640-1820 In. - Lb.
Flexible Drive	Required

TABLE II

SELF-COOLED GENERATOR RATING CHART			
ALTITUDE FEET	INLET AIR TEMP. DEGREES C	GENERATOR MAXIMUM CONTINUOUS LOAD-KVA	
		.75 POWER FACTOR	.985 POWER FACTOR
O-Sea Level	40	42	45
O-Sea Level	20	45	48
O-Sea Level	10	48	48
O-Sea Level	0	48	51
O-Sea Level	-20	51	51
15,000	20	33	36
15,000	10	34	37
15,000	0	36	39
15,000	-20	38	41
25,000	20	24	27
25,000	10	25	29
25,000	0	28	31
25,000	-20	29	34

ITEM		PART NUMBER
1/	Complete System	M21480/18-1
	Generator	M21480/18-2
	Control Panel	MS90313-4
	Current Transformer	M21480/12-6
	Current Transformer	M21480/12-5

1/ The Complete System Consists of: One Generator M21480/18-2, One Control Panel MS90313-4, One Current Transformer M21480/12-6, Three Current Transformers M21480/12-5.

Preparing Activity
Navy - AS
(Project No. 6115-N359)

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